JANNATUL AKIBA

Full Stack Developer | Machine Learning

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github , Linkedin

PROFESSIONAL SUMMARY

Final-year Computer Science and Engineering student at PSTU with project experience across the stack: responsive web applications (React, Next.js), cross-platform mobile apps (Flutter, Firebase), IoT healthcare hardware (Raspberry Pi), and deep learning for medical imaging. Completed Programming Hero's Complete Web Development Course (Batch 14). Seeking a Junior / Trainee Full Stack Developer role or research-oriented position where I can apply engineering fundamentals and grow through real-world delivery.

EDUCATION

B.Sc. in Computer Science and Engineering 2023 – Present

Patuakhali Science and Technology University (PSTU)

RESEARCH EXPERIENCE

Alzheimer's Disease MRI Classification — CSE-410 Thesis *IEEE conference paper in preparation*

- Developed a hybrid InceptionResNetV2 + Vision Transformer model for 4-class dementia staging; eliminated data leakage by evaluating on the original dataset only.
- Applied knowledge distillation to compress the model into MobileNetV2 / EfficientNetV2B0 using a custom Keras Distiller; benchmarked parameters, FLOPs, and latency.
- Compared a frozen Dual-CNN ensemble (92.81%) with fully fine-tuned InceptionResNetV2 (98.67%) and ResNet50 (95.78%).
- Implemented Grad-CAM explainability, ablation studies, a TPU training pipeline, and a Streamlit proof-of-concept interface.

PROJECTS

MediLink Pi — Healthcare Monitoring System (CCE-322) *Hardware + Flutter*

- Built a Raspberry Pi 3B monitoring platform integrating MAX30102, AD8232 (ECG), ADS1115, GSR, and DS18B20 sensors with a Python backend and Flutter dashboard.

CareHealth App — Health Management App Live Link | GitHub

- Developed a cross-platform app (Android and Web) in Flutter with Firebase Auth, Cloud Firestore, push notifications, and Provider state management.

Bazar Dor Monitoring — Market Price Tracker Live Link | GitHub

- Built a responsive Bangla price-tracking web app with live ticker, sortable category pages, and market-wise price tables.
- Integrated Better Auth with email/password, Google, and GitHub sign-in; added profile management, toasts, and skeleton loaders.

FitLog App — Workout Planner Live Link | GitHub

- Created a Next.js / Tailwind CSS app with daily plan, saved workouts, completion tracking, and live calorie and duration totals in a responsive dark UI.

TECHNICAL SKILLS

Programming Language : Python, JavaScript (ES6+), TypeScript, Dart, HTML5, CSS3, LaTeX

Web Development : React.js, Next.js, Tailwind CSS, Bootstrap, Redux, Node.js, Express.js, REST API design, Responsive Design

Mobile app : Flutter, Provider, fl_chart, TableCalendar

Backend & Cloud : Firebase (Auth, Firestore, Cloud Messaging, Hosting), Better Auth, OAuth (Google, GitHub), JWT, MongoDB, Mongoose, MySQL

Machine Learning : TensorFlow/Keras, CNNs, Vision Transformers, Transfer Learning, Knowledge Distillation, Grad-CAM, TPU Training, Streamlit

Embedded & IoT: Raspberry Pi, Sensor Interfacing (MAX30102, AD8232, ADS1115, GSR, DS18B20), Python Backend

Core Engineering: Operating Systems, Computer Networks (Routing & Switching), Simulation & Modeling, Peripheral Interfacing

Tools & Deployment : Git & GitHub, Postman, Vercel, Firebase Hosting, GitHub Pages, Kaggle, VS Code

CERTIFICATIONS & TRAINING

Complete Web Development Course — Programming Hero Batch 14, 2026